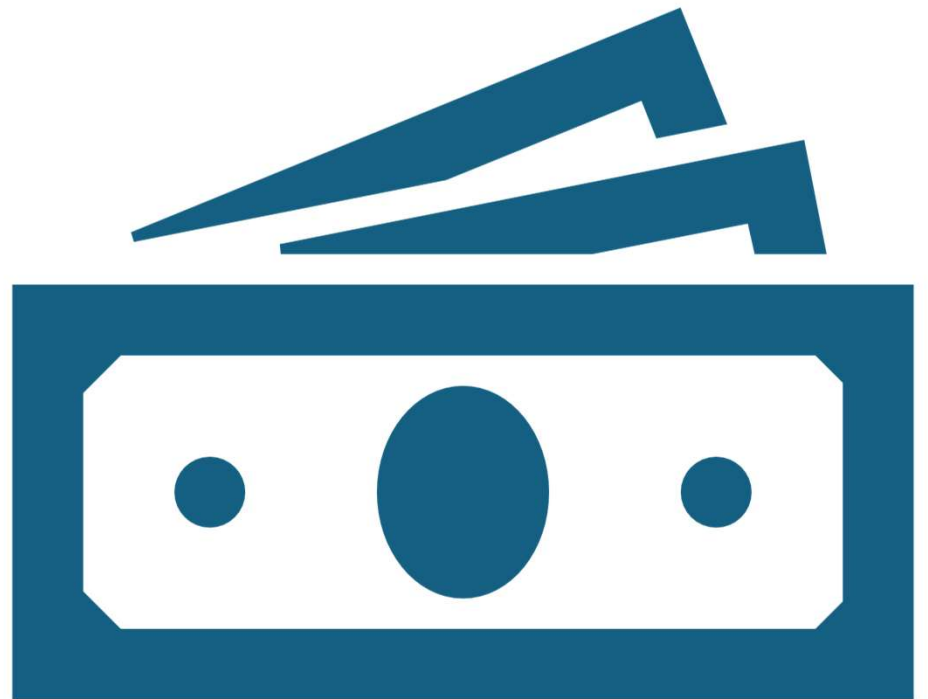


Predicting Movie Investment Risk

Data Science Capstone
Project August 2025

By Jeremy Jex



The Problem

When investing in a movie, how do you know if it will be successful or a flop?

Who Might Care



Movie
Production
companies

Actors

Theaters

Features that can Affect Movies

Movie creators want to profit off their movies and many factors influence that

This data set includes

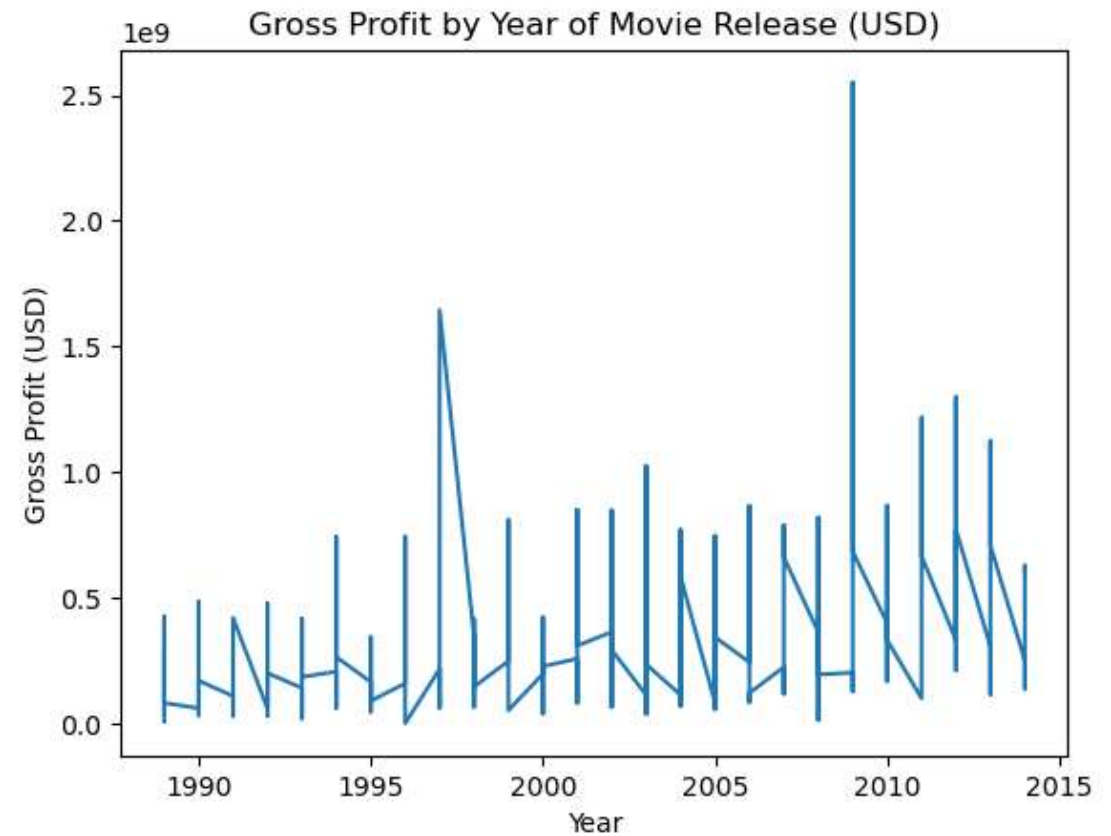
- Title
- Release dates
- Budget
- Gross
- Genre
- Runtime
- MPAA rating
- Review ratings
- Number of Reviews

The Data Set

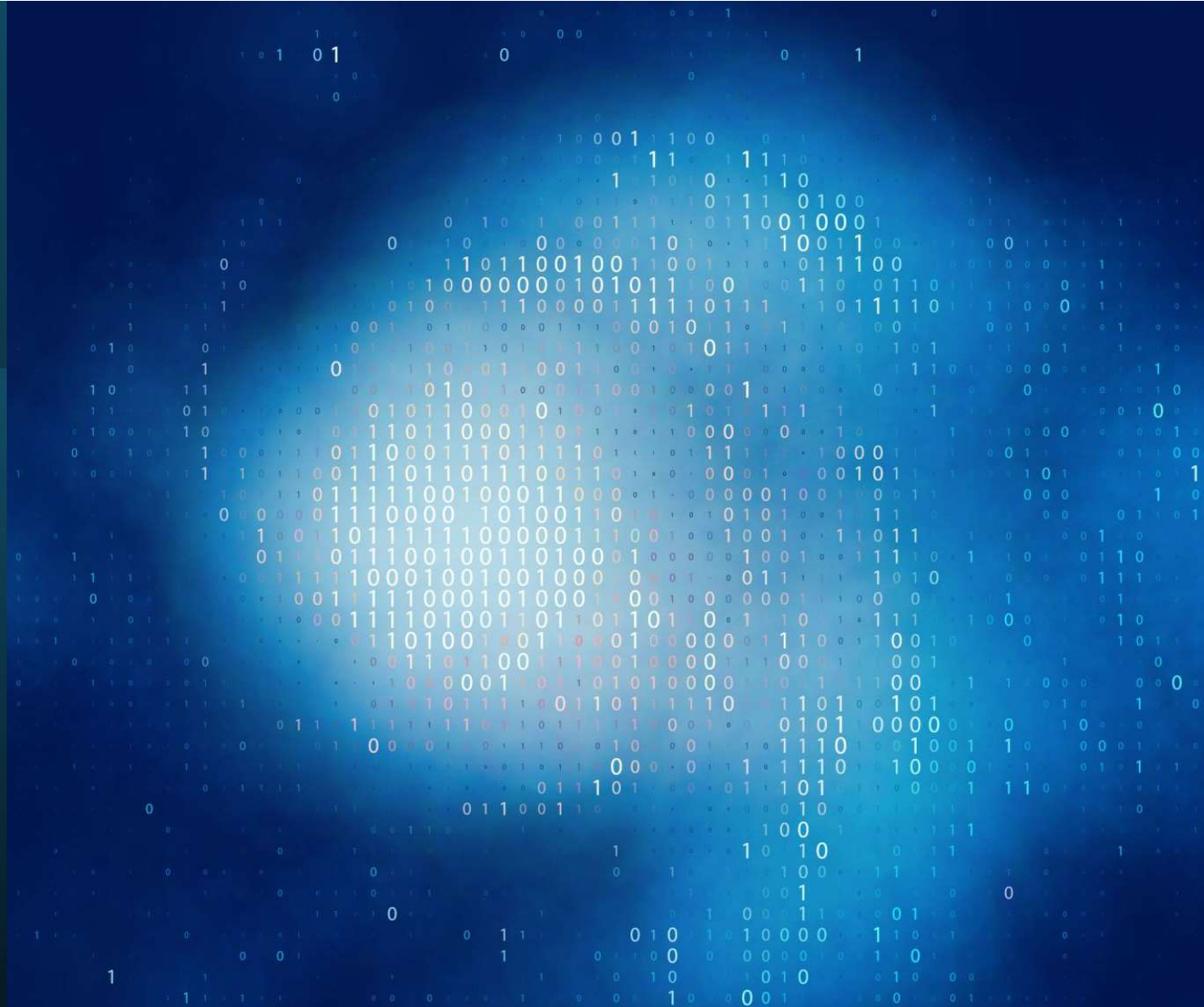
Kaggle Movie data
set

500 Blockbuster
Movies

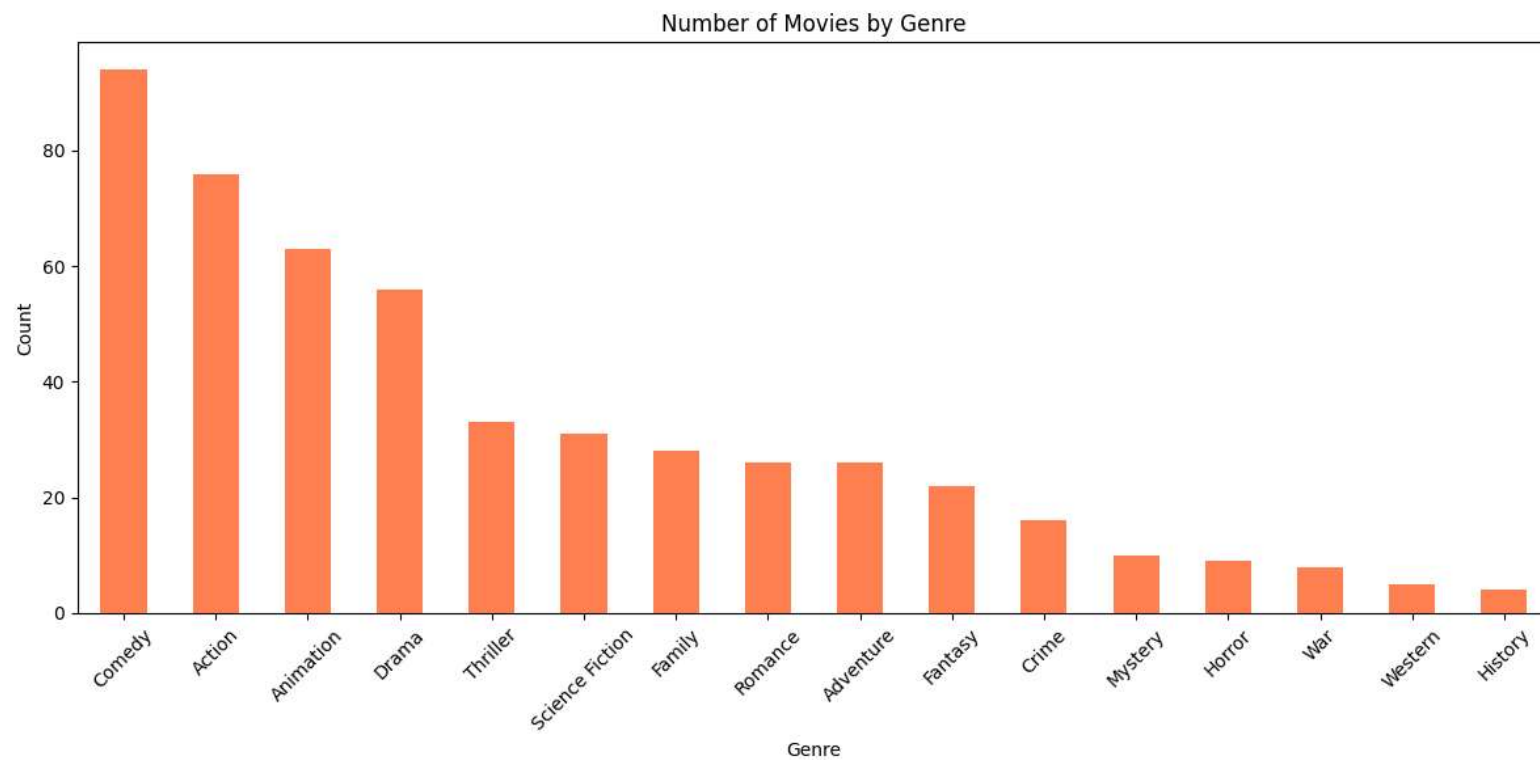
1988 to 2014



Data exploration

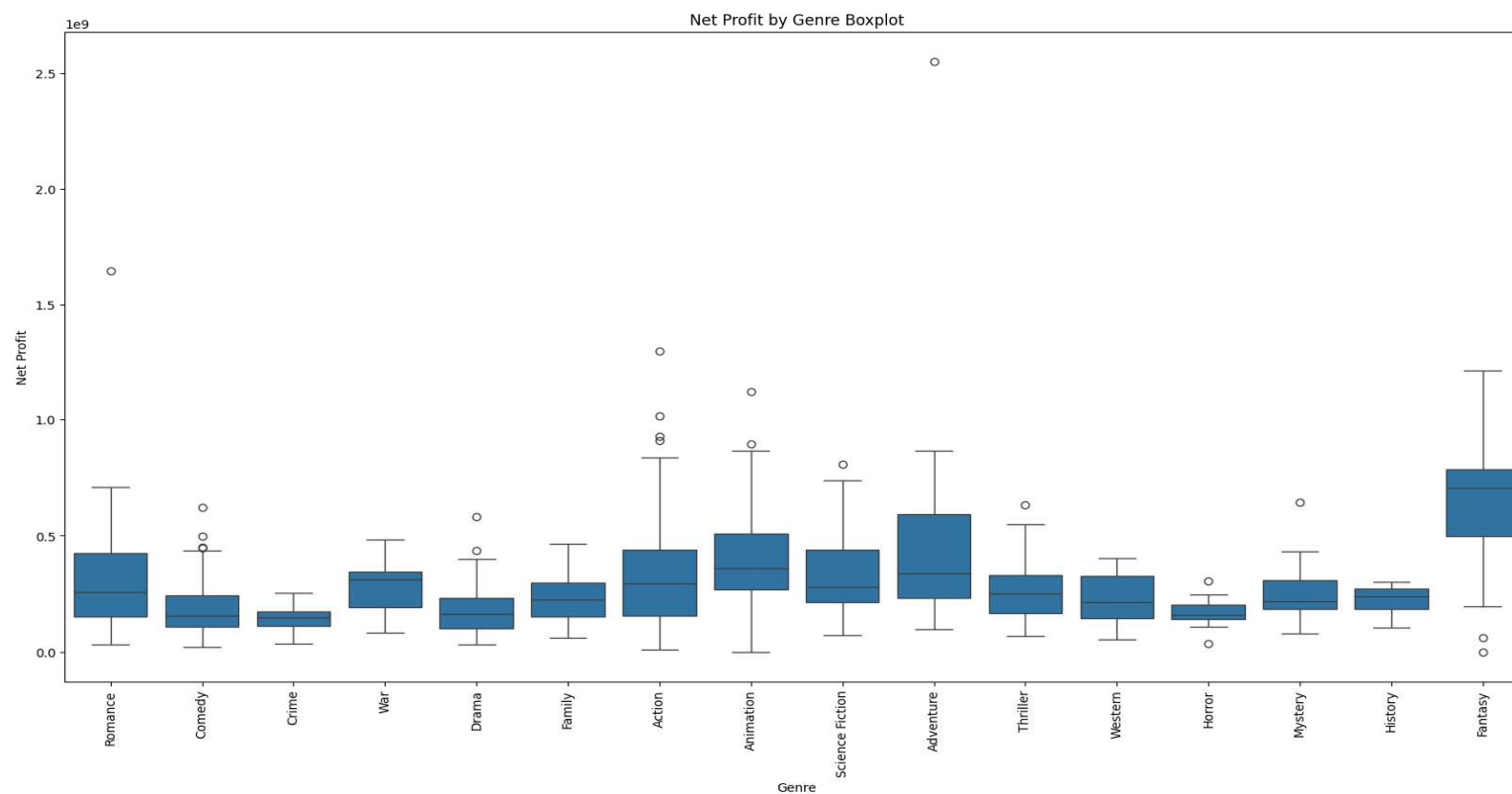


Movies by Genre



Net Profit Grouped by Genre

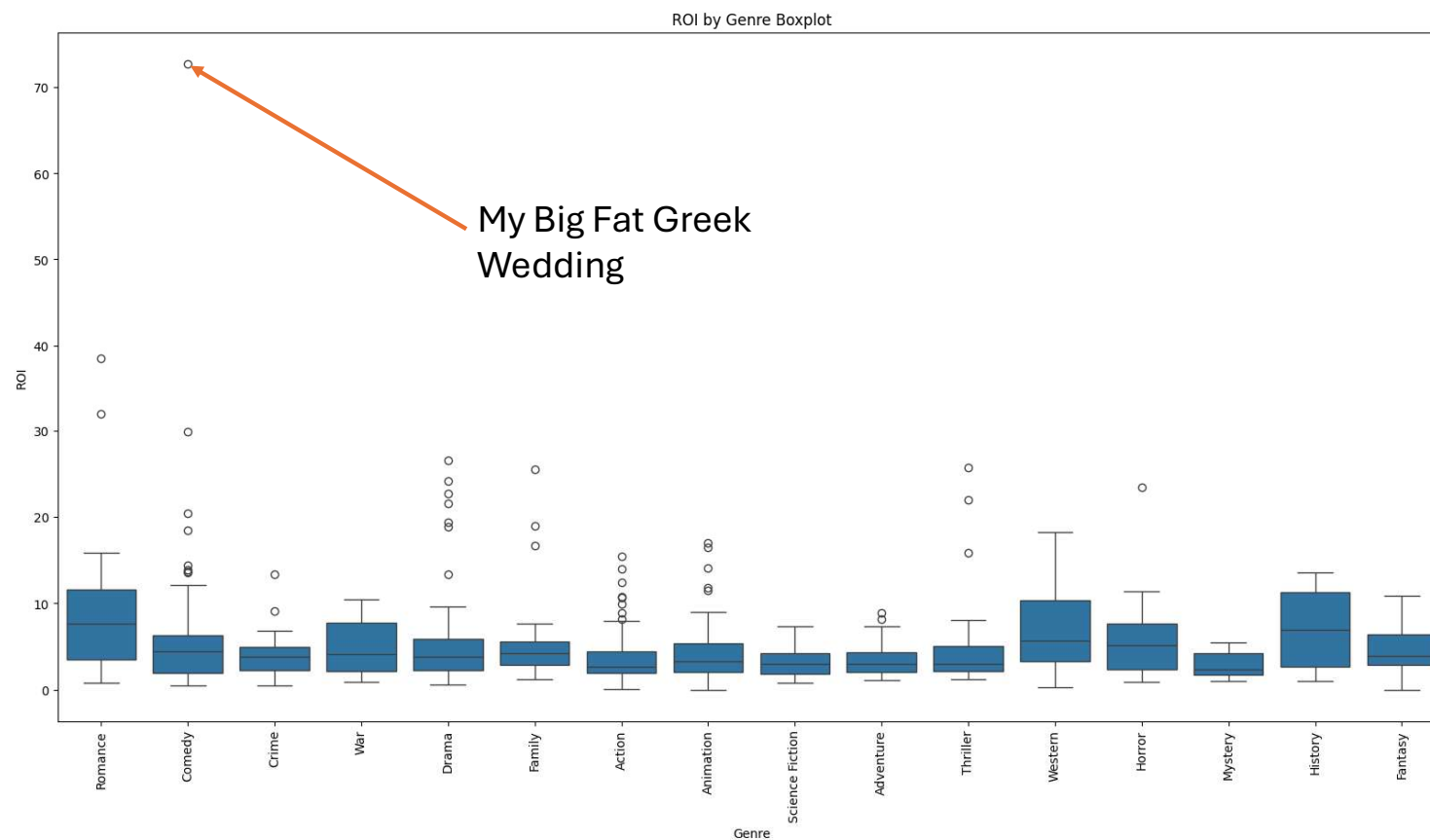
Avatar and Titanic were removed



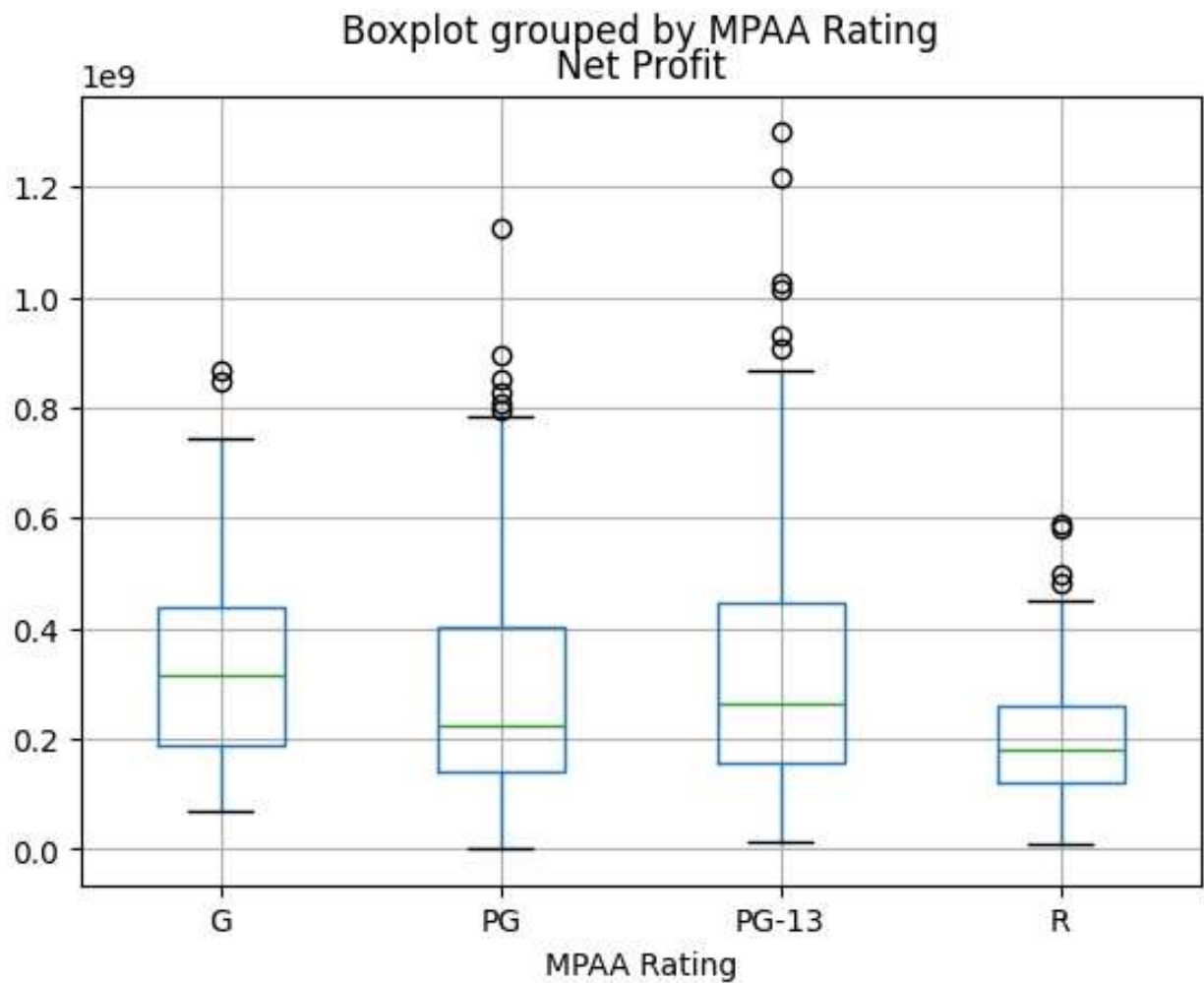
ROI by Genre

Blair Witch Project removed
as outlier

ROI = 4132



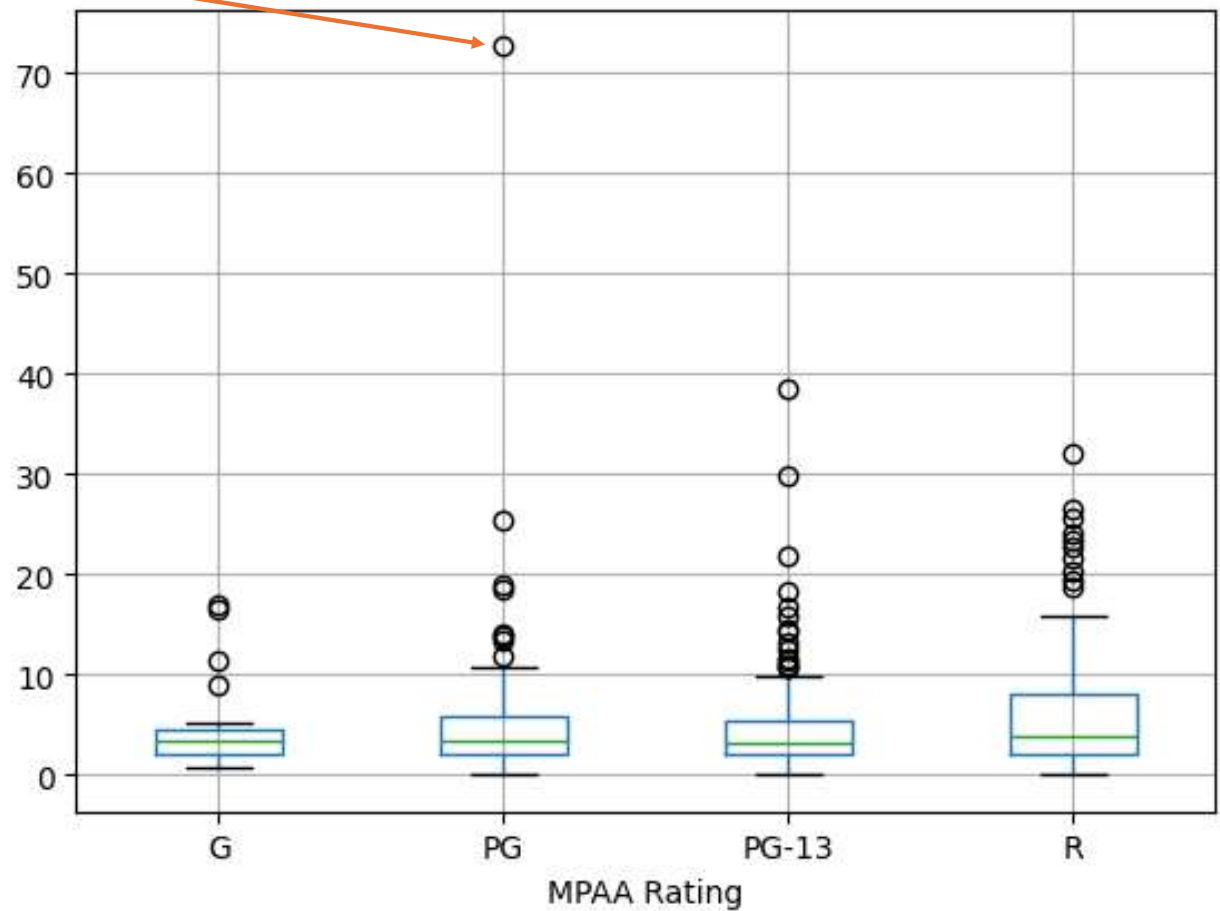
Net Profit grouped by MPAA Rating



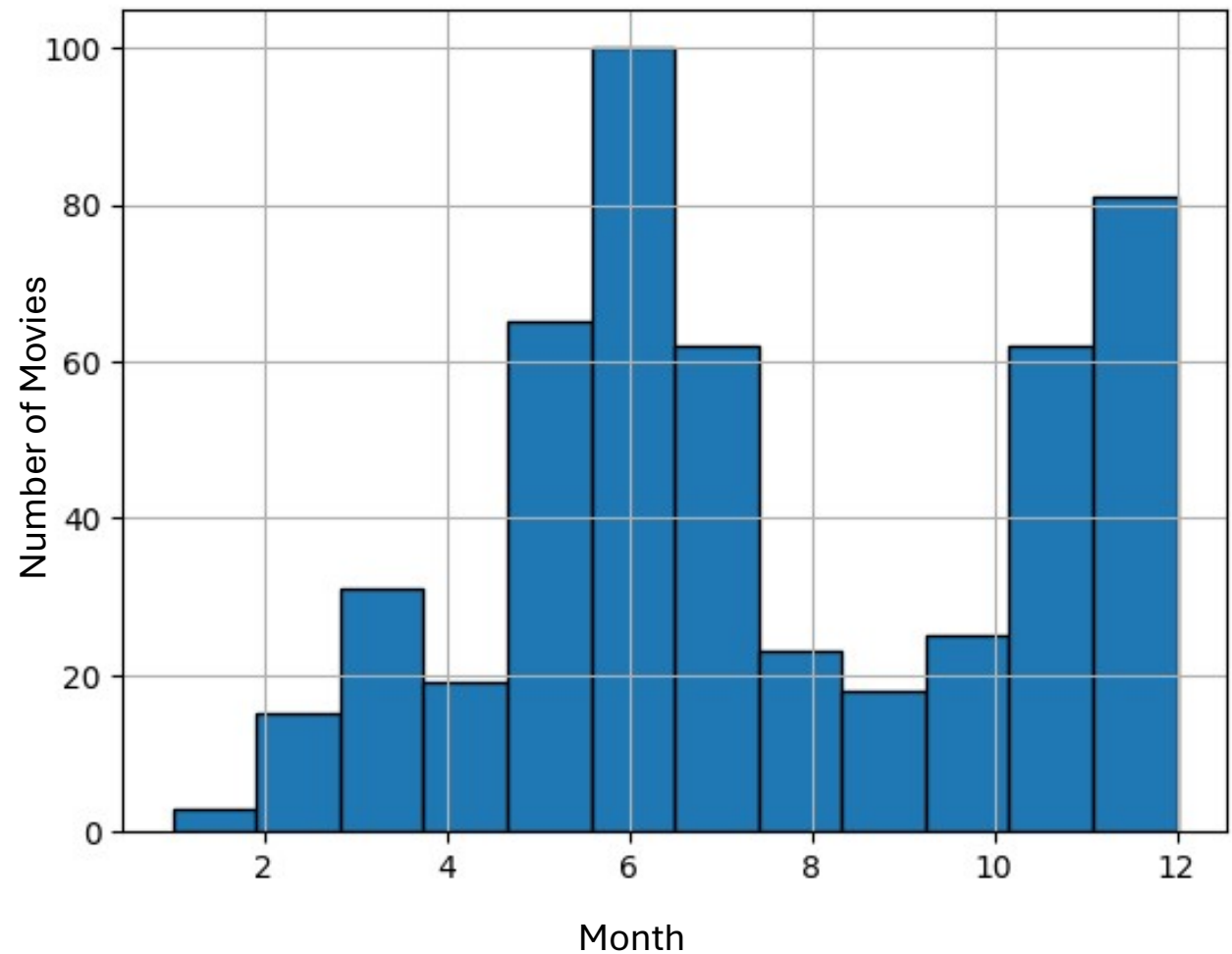
ROI grouped by MPAA Rating

My Big Fat Greek
Wedding

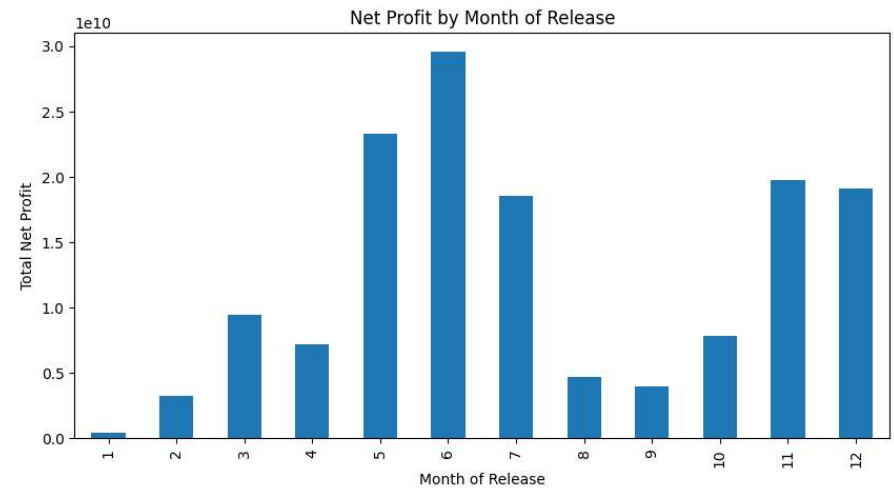
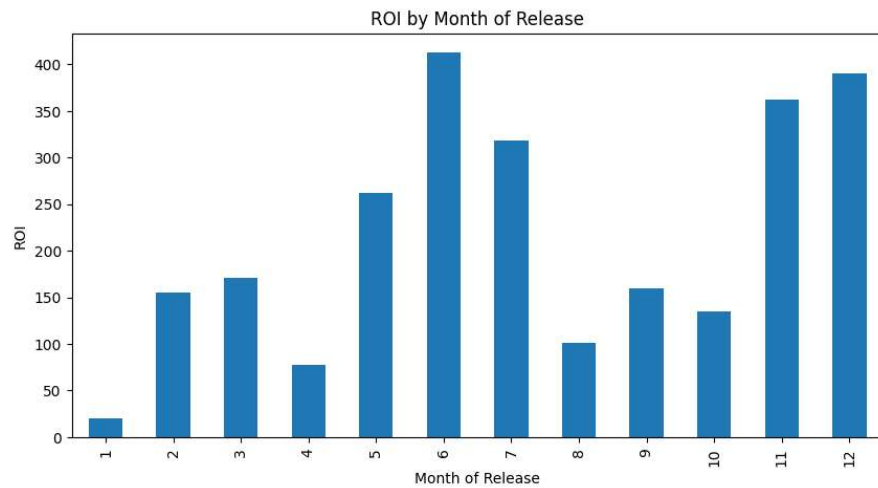
Boxplot grouped by MPAA Rating
ROI



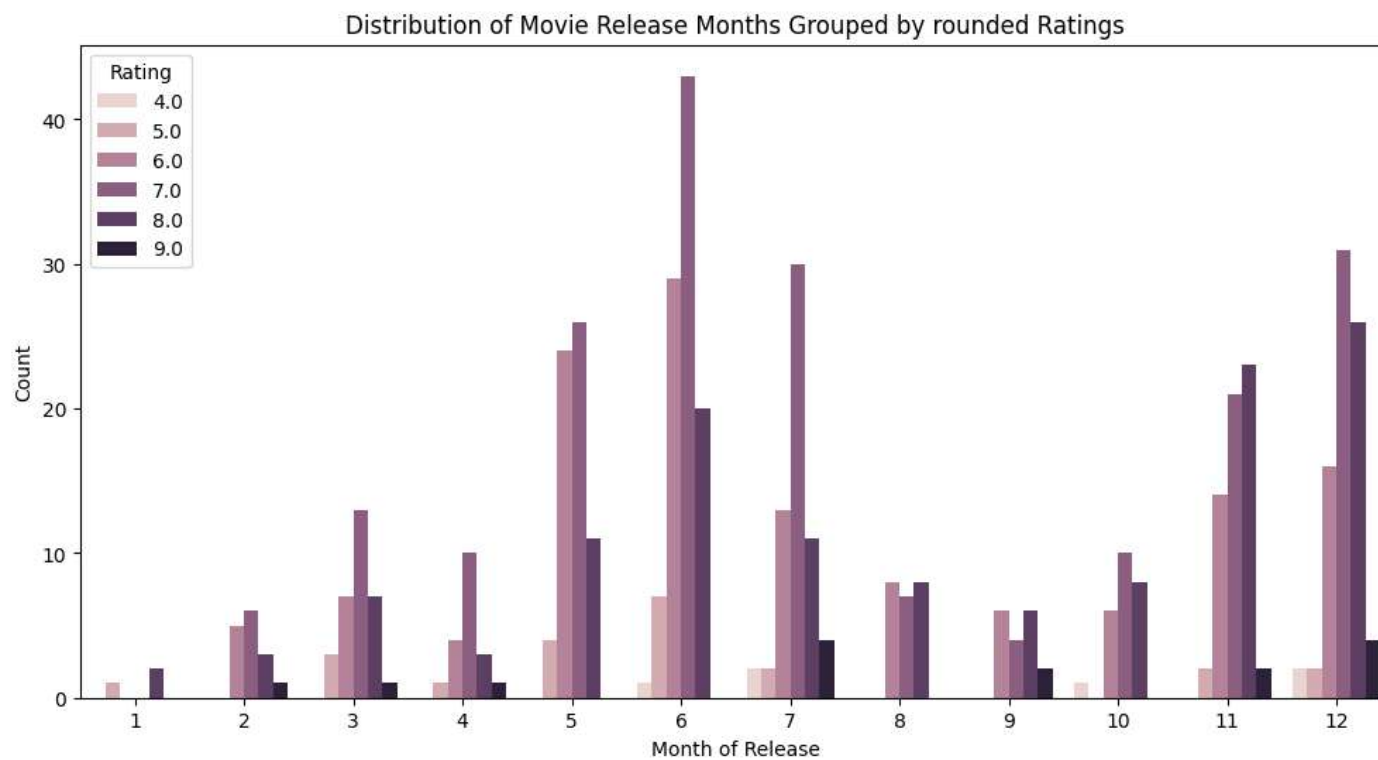
Movies Grouped by Month of Release



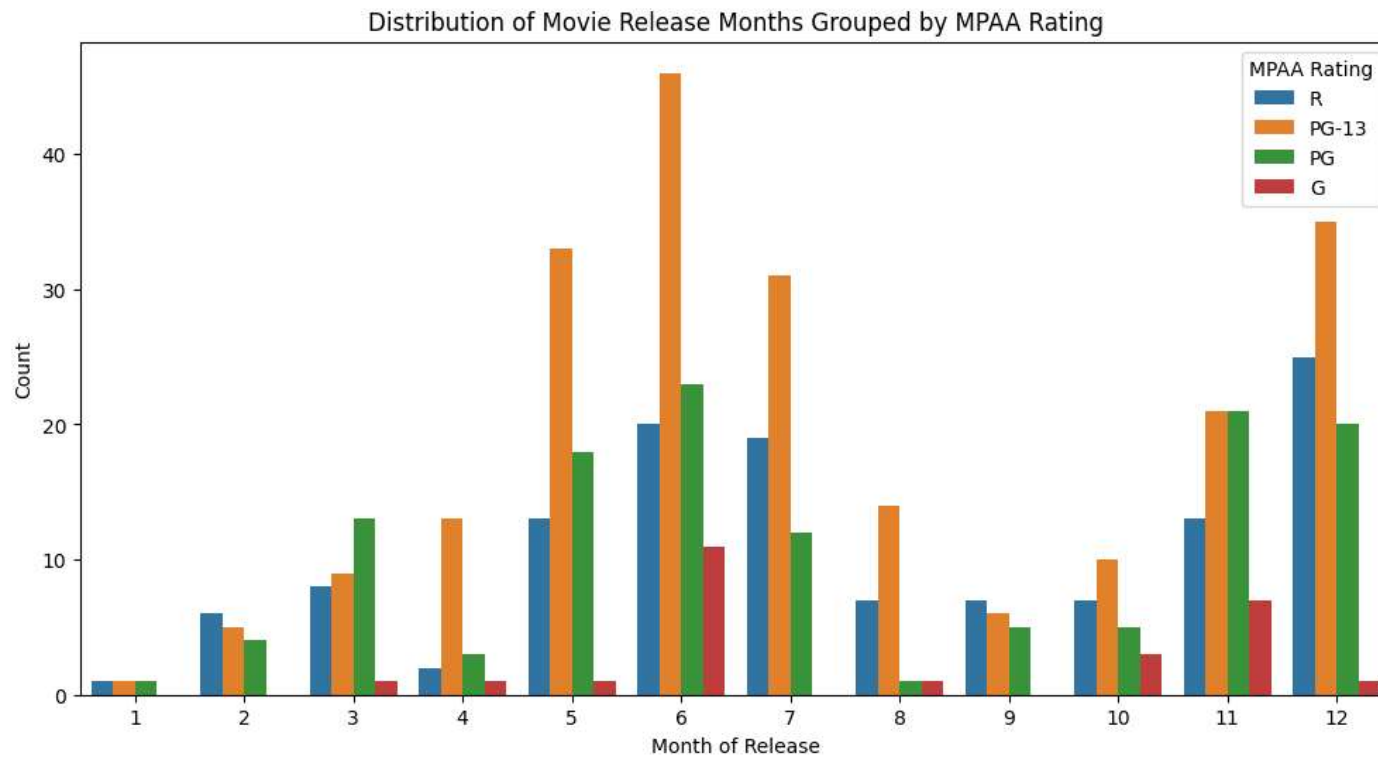
Release Date Grouped by Profits



Release Date grouped by Ratings



Release Date Grouped by MPAA Rating



Heat Map Correlations



Modeling

There were three predictive variables, Gross, ROI and Net Profit.
The Log of the Gross was chosen to be used for modeling

The month was encoded as a sine/cosine feature.

Linear, XGBoost, Random Forest, and SVR was testing for modeling

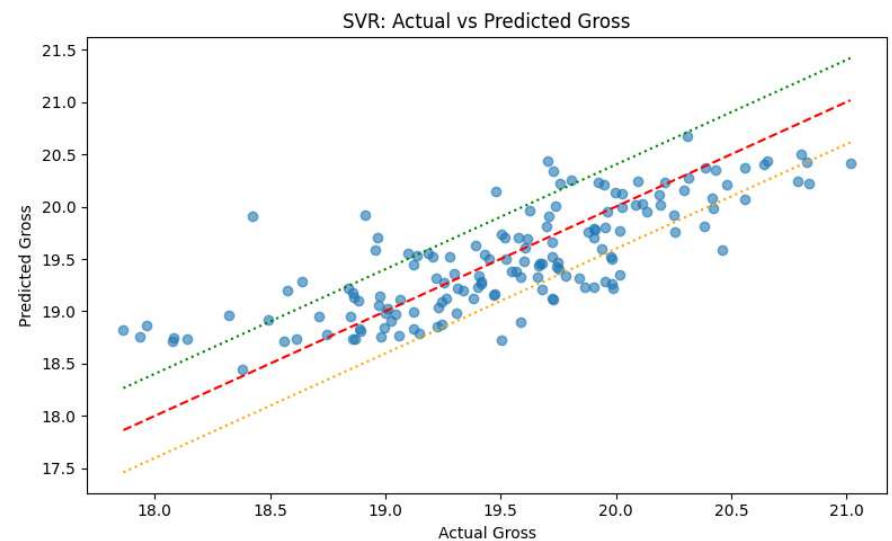
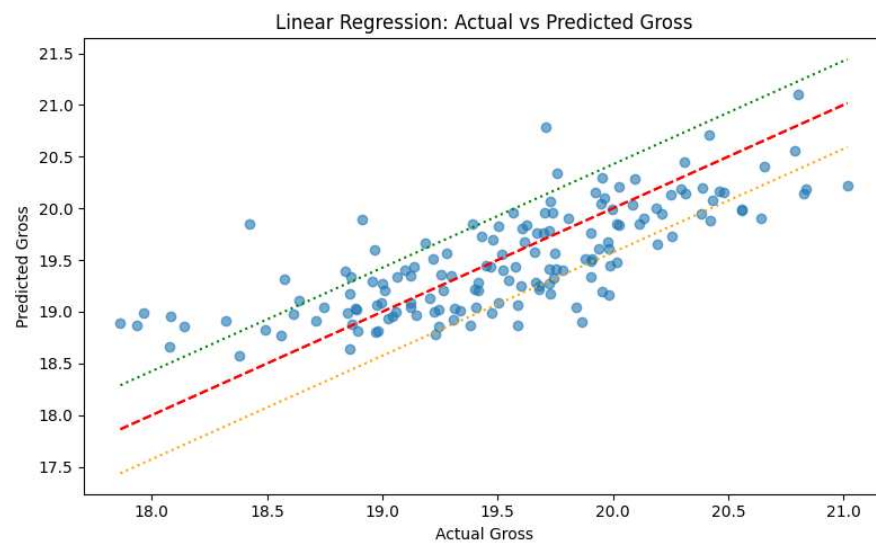
Model metrics

Model	Training R ²	Testing R ²	MAE	K Fold R ²
Random Forest	0.9462	0.5729	0.3217	0.6038
XGBoost	0.9462	0.516	0.337	0.5425
Linear Regression	0.6082	0.5592	0.3442	0.5333
SVR	0.8136	0.6063	0.3159	0.5837
SVR grid search		0.6321	0.4211	

SVR and Linear Regression Graphs

Visually SVR and Linear Regression do not appear that different.

Linear regression processes faster.



Conclusion

- This data set is of Hollywood blockbusters. It shows that successful movies have more critic reviews, higher ratings, and higher budgets.
 - Lower budget movies always have the best ROI and less risk
- Movies released in the June-August or November-December are more successful.
- The final model SVR
 - R squared value = 0.63
 - MAE = 0.42
 - That number looks small but represents millions of dollars.
 - Not the most helpful model
- I think the most helpful information from this project is correlations between release month, review ratings and genres.
- You can't predict if a movie will flop if the data set has no bad movies

Future Study

Should include

- larger sample size
- more robust data set
 - movie flops and blockbuster hits.

Future features

- Directors
- Production companies
- Production time
- Cast
- ect.